INTEGRATION (ANTIDERIVATIVES)

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Lecture Outline

1. Indefinite Integrals
2. Definite Integrals
3. Techniques of Integration
   - Integration by Substitution
   - Integration by Partial Fractions
   - Integration by Parts
4. Reduction formula
5. Improper Integrals



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Antiderivative or Indefinite Integral

Antiderivative is the reverse of derivative.

Definition

Let F and f be two functions such that

$$\frac{\mathrm{d}F(x)}{\mathrm{d}x} = f(x) \text{ for all } x \in I.$$

- f is the derivative of F.
- F is an anti-derivative of $f(x)$ on the interval $I = [a, b]$.

We write

$$F(x) = \int f(x)\mathrm{d}x$$

and read **indefinite integral** of f.

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We write

$$F(x) = \int f(x)\mathrm{d}x$$

and read **indefinite integral** of f. The function $f(x)$ is then called the **integrand**.

Indefinite Integrals

Example

The function $F(x) = \frac{1}{6}x^6$ is an antiderivative of $f(x) = x^5$ since $F'(x) = f(x)$.



Indefinite Integrals

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Remark

If $F(x)$ is an antiderivative of $f(x)$, then all functions $F(x) + c$ where c is a constant is an antiderivative of $f(x)$.



Summary of Some Indefinite Integral (+ c for each case)

1. $\int k\,dx = kx + c$
2. $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + c$



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9. $\int \csc x\,dx = \ln|\csc x - \cot x|$

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Properties of antiderivatives

Given the function f and g, and any constant k

1. $\int k\,\mathrm{d}x = kx + c$
2. $\int \big(f(x) \pm g(x)\big)\,\mathrm{d}x = \int f(x)\mathrm{d}x \pm \int g(x)\mathrm{d}x$
3. $\int kf(x)\mathrm{d}x = k\int f(x)\mathrm{d}x$



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Example

Find $f(x)$ given that

$$f'(x) = 4x^3 - 9 + 2\sin x + 7e^x, \qquad f(0) = 15$$

$$f(x) = \int f'(x)\,dx \tag{1}$$

$$= \int (4x^3 - 9 + 2\sin x + 7e^x)\,dx \tag{2}$$



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Now we plug in $x = 0$ to determine the value of the constant c.

$$f(0) = -2 + 7 + c \tag{4}$$
$$15 = 5 + c \tag{5}$$
$$10 = c \tag{6}$$



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Therefore

$$f(x) = x^4 - 9x - 2\cos x + 7e^x + 10 \tag{7}$$



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Definite Integral

Theorem (Fundamental Theorem of Calculus (1))

$\int_a^b f(x)\mathrm{d}x = F(x)\Big|_a^b = F(b) - F(a)$ *is the definite integral from* a *to* b *of* $f(x)$ *with respect to* x.

Theorem (Integrability of continuous function)

If f *is a continuous function over an interval* $[a, b]$, *or if* f *has at most finitely many jump discontinuities there, then the definite integral* $\int_a^x f(u)\,\mathrm{d}u,\ x \in [a, b]$, *exists and* f *is said to be* **integrable** *over* $[a, b]$.



Properties of Definite Integrals

If f and g are integrable over the interval $[a, c]$, then

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6. If $f(x) \leq g(x), \forall x \in [a, b]$, then $\int_a^b f(x)\ \mathrm{d}x \leq \int_a^b g(x)\ \mathrm{d}x$



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6. If $f(x) \leq g(x), \forall x \in [a, b]$, then $\int_a^b f(x)\ \mathrm{d}x \leq \int_a^b g(x)\ \mathrm{d}x$
7. If $m \leq f(x) \leq M, \forall x \in [a, b]$ then $m \leq \frac{1}{b-a}\int_a^b f(x)\ \mathrm{d}x \leq M$



Theorem

Suppose $f(x)$ *is continuous on* $[-a, a]$ *and*

1. $f(x)$ *is even, then* $\int_{-a}^{a} f(x)\mathrm{d}x = 2\int_{0}^{a} f(x)\mathrm{d}x$
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Example

1. $\int_{-3}^{3} x^2\,dx = 2\int_{0}^{3} x^2\,dx = 2\cdot\left.\frac{x^3}{3}\right|_{0}^{3} = 18$

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2. $\int_{-3}^{3} x^3 dx = 0$
3. $\int_{1}^{x^2} \cos t\,\mathrm{d}t = \int_{1}^{x^2} \cos t\,\mathrm{d}t = \sin t\Big|_1^{x^2} = \sin x^2 - \sin 1$

Fundamental Theorem of Calculus

Theorem (Fundamental Theorem of Calculus (2))

If f *is continuous on* $[a,b]$, *then* $F(x) = \int_a^x f(t)\,\mathrm{d}t$ *is continuous on* $[a,b]$, *differentiable on* (a,b), *and*

$$\frac{\mathrm{d}F(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x).$$



Example

Find $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_5^{2x-1} 3t\sin t\,\mathrm{d}t.$

If $F(x) = \displaystyle\int_5^{2x-1} 3t\sin t\,\mathrm{d}t,$



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If $F(x)=\displaystyle\int_5^{2x-1} 3t\sin t\,\mathrm{d}t,$ then for $u=2x-1,$



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$$=6(2x-1)\sin(2x-1) \tag{10}$$



Outline of Presentation

1. Indefinite Integrals
2. Definite Integrals
3. Techniques of Integration
   - Integration by Substitution
   - Integration by Partial Fractions
   - Integration by Parts
4. Reduction formula
5. Improper Integrals



Techniques of integration.

Sometimes it is not easy to guess $F(x)$. Then some integration techniques could be employed to find $F(x)$.
The following techniques are discussed:

1. Integration by substitution
2. Integration by Parts
3. Integration by Partial fraction



Integration by substitution

Here we introduce a new variable to transform the original problem into one that can be easily solved.

Example

Evaluate $\int 2x\sqrt{1+x^2}\,\mathrm{d}x$.



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$$\begin{aligned}\int 2x\sqrt{1+x^2}\,\mathrm{d}x &= \int 2x\sqrt{u}\cdot\frac{\mathrm{d}u}{2x}\\ &= \int \sqrt{u}\,\mathrm{d}u\\ &= \frac{2}{3}u^{3/2}+c = \frac{2}{3}(1+x^2)^{3/2}+c\end{aligned}$$



Example

Find $\int 7(2x-1)(x^2-x+3)^4 \mathrm{d}x$.



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$$= \frac{7}{5}(x^2-x+3)^5 \tag{13}$$



Integration using partial fractions

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$$\frac{2}{x-1}+\frac{5}{x+4}=$$



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4. Partial fraction decomposition can sometimes be used to simplify integrands to make integrations easy to carry out.



Integration using partial fractions

Recall that a rational function of the form $f(x) = \dfrac{ax+b}{(px+q)(rx+s)}$ decomposes by

$$\frac{ax+b}{(px+q)(rx+s)} = \frac{A}{(px+q)} + \frac{B}{(rx+s)}$$



Example

Evaluate $\int \frac{7x+3}{(x-1)(x+4)}\mathrm{d}x.$



Example

Evaluate $\int \frac{7x+3}{(x-1)(x+4)}\mathrm{d}x$.

$$\begin{aligned}\int \frac{7x+3}{(x-1)(x+4)}\mathrm{d}x &= \int \frac{2}{x-1}+\frac{5}{x+4}\mathrm{d}x \\ &= \int \frac{2}{x-1}\mathrm{d}x+\int \frac{5}{x+4}\mathrm{d}x \\ &= 2\ln|x-1|+5\ln|x+4|+c\end{aligned}$$



Integration using partial fractions

Also a rational function of the form $f(x) = \dfrac{ax+b}{(px+q)^2}$ decomposes into

$$\frac{ax+b}{(px+q)^2} = \frac{A}{(px+q)} + \frac{B}{(px+q)^2}$$

Example

Evaluate $\displaystyle\int \frac{10x+18}{(2x+3)^2}\mathrm{d}x.$



1. We have that $\dfrac{10x+18}{(2x+3)^2} = \dfrac{A}{(2x+3)} + \dfrac{B}{(2x+3)^2} = \dfrac{A(2x+3)+B}{(2x+3)^2}$



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$$\begin{aligned}\int \frac{10x+18}{(2x+3)^2}\mathrm{d}x &= \int \frac{5}{(2x+3)} + \frac{3}{(2x+3)^2}\mathrm{d}x \\ &= \int \frac{5}{(2x+3)}\mathrm{d}x + \int \frac{3}{(2x+3)^2}\mathrm{d}x \\ &= \frac{5}{2}\ln|2x+3| - \frac{3}{2(2x+3)} + c\end{aligned}$$



Integration by parts

Theorem (Integration by Parts)

If u and v are two continuous functions on the interval $[a,b]$, then

$$\int u'v\,\mathrm{d}x = \int (uv)'\,\mathrm{d}x - \int uv'\,\mathrm{d}x = uv - \int uv'\,\mathrm{d}x \tag{14}$$

or

$$\int_a^b u'v\,\mathrm{d}x = \int_a^b (uv)'\,\mathrm{d}x - \int_a^b uv'\,\mathrm{d}x = uv\Big|_a^b - \int_a^b uv'\,\mathrm{d}x \tag{15}$$



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Proof.

Indeed, the product rule $(uv)' = u'v + uv'$ implies $u'v = (uv)' - uv'$.

That is $\int_a^b u'v(x)\,\mathrm{d}x = uv\Big|_a^b - \int_a^b uv'\,\mathrm{d}x$. □

Example

Evaluate $\int x \cos x \mathrm{d}x$

1. Let $u'(x) = \cos x$ and $v(x) = x$.



Example

Evaluate $\int x \cos x \mathrm{d}x$

1. Let $u'(x) = \cos x$ and $v(x) = x$.
2. We have $u(x) = \sin x$ and $v'(x) = 1$.



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$$= x\sin x + \cos x + c \tag{18}$$



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Remark

Note that we can also let $u'(x)=x$ and $v(x)=\cos x$; however, this will make the task more difficult than before.

The choice depends on which function you can easily integrate or differentiate.

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Evaluate $\int_0^1 xe^x \mathrm{d}x$

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3. Implying $u(x) = e^x$, and $v'(x) = -\sin x$



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$$F(x) = e^x \cos x + \int e^x \sin x \mathrm{d}x \tag{23}$$



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$$F(x) = e^x \cos x + \left(e^x \sin x - \int e^x \cos x \mathrm{d}x \right) \tag{24}$$



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Reduction formula

Sometimes in the process of integration by parts, one has to perform the process severally to arrive at the desired solution. This often can be shortened by identifying some patterns which can be used to derive formulae to complete the computation.



Reduction formula

Sometimes in the process of integration by parts, one has to perform the process severally to arrive at the desired solution. This often can be shortened by identifying some patterns which can be used to derive formulae to complete the computation. For example that given that

$$R_n = \int x^n e^x \mathrm{d}x$$

then by integration by parts we have

$$\begin{aligned} R_n &= x^n e^x - n\int x^{n-1} e^x \mathrm{d}x \\ &= x^n e^x - nR_{n-1} \end{aligned}$$



Reduction formula

Sometimes in the process of integration by parts, one has to perform the process severally to arrive at the desired solution. This often can be shortened by identifying some patterns which can be used to derive formulae to complete the computation. For example that given that

$$R_n = \int x^n e^x \mathrm{d}x$$

then by integration by parts we have

$$\begin{aligned} R_n &= x^n e^x - n\int x^{n-1} e^x \mathrm{d}x \\ &= x^n e^x - nR_{n-1} \end{aligned}$$

Notice that R_{n-1} follows the same procedure as done for R_n. Thus we have derived a reduction formula $R_n = x^n e^x - nR_{n-1}$ which holds for all n.



Example

Evaluate $\int x^3 e^x \mathrm{d}x$.



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&= x^3 e^x - 3\left(x^2 e^x - 2\left[x e^x - \int e^x \mathrm{d}x\right]\right) \\
&= x^3 e^x - 3\left(x^2 e^x - 2\left[x e^x - e^x\right]\right) + c
\end{aligned}$$



Let now turn to the reduction formula for the $\tan$ function. For example given that

$$u_n = \int \tan^n x \mathrm{d}x \tag{28}$$

for $n \geq 1$, then we have

$$\int \tan^n x \mathrm{d}x = \int \tan^{n-2} x \tan^2 x \mathrm{d}x \tag{29}$$



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Therefore reduction formula is

$$u_n = \frac{1}{n-1} \tan^{n-1} x - u_{n-2} \tag{33}$$



Example

Evaluate $\int \tan^5 x \mathrm{d}x$.



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1. We now apply the formula to find $u_5 = \int \tan^5 x \mathrm{d}x$. This is given by

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$$= \frac{1}{4}\tan^4 x - u_3 \tag{35}$$



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This means

$$u_5 = \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \ln|\cos x| + c \tag{38}$$



Outline of Presentation

1. Indefinite Integrals
2. Definite Integrals
3. Techniques of Integration
   - Integration by Substitution
   - Integration by Partial Fractions
   - Integration by Parts
4. Reduction formula
5. Improper Integrals



Improper Integrals of Type I

Definition

Integrals $\int_a^b$ with $a = -\infty$ or $b = +\infty$ are improper integrals of type I.



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3. $\int_{-\infty}^{+\infty} f(x)\mathrm{d}x = \int_{-\infty}^c f(x)\mathrm{d}x + \int_c^{+\infty} f(x)\mathrm{d}x$ *if* f *is continuous on* $(-\infty, +\infty)$ *and* c *is any real number.*

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Definition

If the limit exists and is finite, we say that the improper integral **converges**. Otherwise, it **diverges**.



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The limit exists and its finite, therefore the improper integral converges.



Improper Integrals of Type II

Definition

Integrals $\int_a^b f(x)\mathrm{d}x$ with $f(a)=\infty$ or $f(b)=\infty$ are improper integrals of type II.

Theorem

1. $\int_a^b f(x)\mathrm{d}x = \lim_{c\to a^+}\int_c^b f(x)\mathrm{d}x$ if f is right-discontinuous at a.

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3. $\int_a^b f(x)\mathrm{d}x = \int_a^c f(x)\mathrm{d}x + \int_c^b f(x)\mathrm{d}x$ if f is discontinuous at c but continuous on $(a,c)\cup(c,b)$.

Example

Evaluate $\int_0^1 \frac{1}{1-x}\mathrm{d}x$



Example

Evaluate $\int_0^1 \frac{1}{1-x}\mathrm{d}x$

$$\int_0^1 \frac{1}{1-x}\mathrm{d}x = \lim_{b\to 1^-}\Big[-\ln|1-x|\Big]_0^b \tag{40}$$

$$= +\infty \tag{41}$$



Exercise

1. Find an anti-derivative of f if

   1) $f(x) = x^2(x^3-1)^8$ 2) $f(x) = \sin^2 x \cos x$ 3) $f(u) = 4\frac{\sin u}{\cos^2 u}$

   4) $f(y) = y\cos y^2$ 5) $f(x) = \frac{3x^2-2x}{\sqrt{1+x^3-x^2}}$ 6) $f(x) = \frac{3x^2-2x}{\sqrt{1-(x^3-x^2)^2}}$

   7) $f(x) = x\,3^{x^2+1}$ 8) $f(s) = \frac{s^2-3s+1}{s+1}$ 9) $f(x) = \frac{4}{(x-2)(-x+4)}$.

2. Find the reduction formula of $\int f_n(x)\mathrm{d}x$ if

   (a) $f_n(x) = \sin^n x$,

   (b) $f_n(x) = \cos^n x$

   (c) $f_n(x) = \ln^n x$



Exercise

1. Find the indefinite integral $F(x) = \int f(x)\mathrm{d}x$ that satisfies $F(a) = A$ if
   $(a)\ f(x) = \ln x, a = 1, A = 0$
   $(b))\ f(x) = x^2 e^x, a = 1, A = e$
   $(c)\ f(x) = (x^2 - 2x)e^{2x}$
   $(d)\ f(x) = x^3 \ln x, a = 1, A = -2$
   $(e)\ f(z) = \mathrm{asin}\, z, a = 0, A = 1/2$
2. Evaluate $\int f$ by using a substitution prior to integration by parts.
   $(a)\ f(s) = e^{\sqrt{3s+9}}$
   $(b)\ f(x) = \ln\left(x + x^2\right)$
   $(c)\ f(y) = \sin\left(\ln y\right)$
   $(d)\ f(z) = z(\ln z)^2$



Exercise

1. Determine whether the following integrals converge or diverge.
   1) $\int \frac{\ln x}{x^2}\mathrm{d}x$ 2) $\int_{-\infty}^{+\infty} \frac{1}{1+x^2}\mathrm{d}x$
   3) $\int_{-\infty}^{-2} \frac{2}{x^2-1}\mathrm{d}x$ 4) $\int_{2}^{+\infty} \frac{2}{x^2-x}\mathrm{d}x$.
2. Evaluate the following $\int_a^b f(x)\mathrm{d}x$ if
   1) $f(x)=\frac{1}{\sqrt{4-x}}, a=0, b=4$ 2) $f(x)=\frac{1}{\sqrt{1-x^2}}, a=0, b=1$
   3) $f(x)=\frac{1}{x\ln x}, a=1, b=2$ 4) $f(x)=\frac{1}{\sqrt{x}+\sin x}, a=0, b=\pi$.
3. Evaluate 1) $\int \frac{9-9x}{(2x-1)(x+4)}\mathrm{d}x$.
   2) $\int \frac{4x^2}{(x-1)(x-2)^2}\mathrm{d}x$.



END OF LECTURE

THANK YOU

